

Assignment 2

Task 1:

$$a) (B \cdot C' + A' \cdot D)(AB' + (C \cdot D)')$$

$$\underbrace{(B \cdot C') \cdot (A \cdot B')}_0 + \underbrace{(B \cdot C') \cdot ((C \cdot D)')}_0 + \underbrace{(A' \cdot D) \cdot (A \cdot B')}_0 + \underbrace{(A' \cdot D) \cdot (C \cdot D)'}_0$$

$$(B \cdot C' + A' \cdot D)(AB' + (C \cdot D)') = \underline{\underline{0}}$$

$$b) (x \cdot y' + x' \cdot y)'$$

$$(x \cdot y')' \cdot (x' \cdot y)'$$

$$(x' + y)(x + y')$$

$$x' \cdot x + x' \cdot y' + y \cdot x + y \cdot y'$$

$$\underline{\underline{x'y' + yx}}$$

Task 2:

$$a) (u + xw) \cdot (x + u' \cdot v)$$

$$ux + \cancel{u(u'v)} + xwx + xw(u'v)$$

$$ux + xw + xwu'v$$

$$F(u, v, w, x) = \underline{\underline{\Sigma(3, 7, 9, 11, 13, 15)}}$$

u	v	w	x	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Task 2:

$$b) x' + x(x+y') \cdot (y+z')$$

$$x' + x(y+z')$$

$$x' + y + z'$$

$$F(x, y, z) = \underline{\underline{\Sigma(0, 1, 2, 3, 4, 6, 7)}}$$

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Task 3:

$$a) F(x, y, z) = \Sigma(1, 2, 3, 6, 7)$$

$$\underline{\underline{F = y + xz}}$$

	$y'z'$	$y'z$	yz	yz'
x'	0	1	1	1
x	0	0	1	1

$$b) F(x, y, z) = x'y + yz' + y'z'$$

	$y'z'$	$y'z$	yz	yz'
x'	1	0	1	1
x	1	0	0	1

$$\underline{\underline{F = z' + x'y}}$$

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Task 4:

a) $F(A, B, C, D) = A'B'C'D' + AC'D' + B'C'D' + A'BCD + BC'D$

	$A'B'$	$A'B$	AB	AB'
$C'D'$	1	0	1	1
$C'D$	0	1	1	0
CD	0	1	0	0
CD'	1	0	0	1

$F = B'D' + ABC' + A'BD$

b) $F(w, x, y, z) = x'z + w'xy' + w(x'y + xy')$

	$w'x'$	$w'x$	wx	wx'
$y'z'$	0	1	1	0
$y'z$	1	1	1	1
yz	1	0	0	1
yz'	0	0	0	1

$F = xy' + zx' + wx'y$

Task 5:

a) $F(A, B, C, D) = \prod(1, 3, 5, 7, 13, 15)$

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

	$A'B'$	$A'B$	AB	AB'
$C'D'$	1	1	1	1
$C'D$	0	0	0	1
CD	0	0	0	1
CD'	1	1	1	1

$F' = A'D + B \cdot D$

$F = (A'D + B \cdot D)'$
 $= (A'D)' \cdot (B \cdot D)'$
 $= (A + D')(B' + D')$

Task 5:

$$b) F(x, y, z) = x'z' + y'z' + yz' + xy \\ = z' + xy$$

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

	$y'z'$	$y'z$	yz	yz'
x'	1	0	0	1
x	1	0	1	1

$$F' = \underline{zx' + y'z}$$

$$F = (zx' + y'z)' \\ = (zx')'(y'z)' \\ = \underline{\underline{(z' + x)(y + z')}}$$

Task 6:

$$a) F(A, B, C, D) = \Sigma(5, 6, 7, 12, 14)$$

$$d(A, B, C, D) = \Sigma(3, 9, 11, 15)$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	x
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	x
1	0	1	0	0
1	0	1	1	x
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	x

	$A'B'$	$A'B$	AB	AB'
$C'D'$	0	0	1	0
$C'D$	0	1	0	x
CD	x	1	x	x
CD'	0	1	1	0

$$F = \underline{\underline{CD + BC + A'BD + ABD'}}$$

Task 6:

$$b) F(A, B, C, D) = \Sigma(2, 4, 7, 10, 12)$$

$$d(A, B, C, D) = \Sigma(0, 6, 8)$$

A	B	C	D	F
0	0	0	0	X
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	X
0	1	1	1	1
1	0	0	0	X
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

	A'B'	A'B	AB	AB'
C'D'	X	1	1	X
C'D	0	0	0	0
CD	0	1	0	0
CD'	1	X	0	1

Already covered by C'D'

$$F = C'D' + B'D' + \cancel{A'C'D'} + A'BC$$

$$\underline{\underline{F = C'D' + B'D' + A'BC}}$$